

Name: Xin Ling Chong

Student id: 35664029

Date: 7/9/2025

UNF:

PURPOSE(purpose_no,purpose_desc,(combination_code,combination_hire_rate,
truck_vin,truck_rego_no,trailer_code,trailer_manufacturer,trailer_model,
trailer_load_capacity,trailer_dimensions))

1NF:

PURPOSE(purpose_no,purpose_desc)

Candidate key:

purpose_no

COMBINATION(purpose_no, combination_code,combination_hire_rate,
truck_vin,truck_rego_no,trailer_code,trailer_manufacturer,trailer_model,
trailer_load_capacity,trailer_dimensions)

Candidate keys:

purpose_no, combination_code

purpose_no, truck_vin, trailer_code

Partial dependencies:

truck_vin -> truck_rego_no

trailer_code -> trailer_manufacturer,trailer_model,trailer_load_capacity,trailer_dimensions

2NF:

PURPOSE(purpose_no,purpose_desc)

COMBINATION(purpose_no, combination_code,combination_hire_rate,truck_vin,
trailer_code)

TRUCK(truck_vin, truck_rego_no)

TRAILER(trailer_code,trailer_manufacturer,trailer_model, trailer_load_capacity,
trailer_dimensions)

No transitive dependencies present

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3NF:

PURPOSE(purpose_no, purpose_desc)

COMBINATION(purpose_no, combination_code, combination_hire_rate, truck_vin, trailer_code)

TRUCK(truck_vin, truck_rego_no)

TRAILER(trailer_code, trailer_manufacturer, trailer_model, trailer_load_capacity, trailer_dimensions)

Full dependencies:

purpose_no -> purpose_desc

purpose_no, combination_code -> combination_hire_rate, truck_vin, trailer_code

truck_vin -> truck_rego_no

trailer_code -> trailer_manufacturer, trailer_model, trailer_load_capacity, trailer_dimensions